

The Hilbert–Pólya Conjecture and Pandrosion Spectral Constraints

Ivan Besevic

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Abstract

The Hilbert–Pólya conjecture posits a self-adjoint operator H whose eigenvalues are the ordinates γ_n of the nontrivial zeros $\rho_n = 1/2 + i\gamma_n$ of the Riemann zeta function. We derive *necessary Pandrosion conditions* that any such operator must satisfy: (1) its resolvent is $G(z) = P'(z)/P(z)$ where $P = \prod (z - \gamma_k)$, (2) the vanishing identity $\sum 1/P'(\gamma_k) = 0$ holds, (3) the Pandrosion zeta $\zeta_P(s) = \sum P'(\gamma_k)^{-s} = 0$ for $s = 1, 2, 3$, (4) the density of states is $\rho = -(1/\pi) \text{Im}(P'/P)$, and (5) the tidal field $T = E_P/P^2$ encodes the pair correlation. All conditions are verified on the first 30 zeros of ζ to the precision 10^{-42} . These provide concrete, falsifiable tests for any candidate Hilbert–Pólya operator.

1 The conjecture

Conjecture 1.1 (Hilbert–Pólya). *There exists a self-adjoint (unbounded) operator H on a Hilbert space such that*

$$\text{Spec}(H) = \{\gamma_n : \zeta(1/2 + i\gamma_n) = 0, \gamma_n > 0\}. \quad (1)$$

If H exists, its characteristic polynomial (in a finite truncation to the first N zeros) is:

$$P(t) = \prod_{n=1}^N (t - \gamma_n), \quad P \in \mathbb{R}[t] \quad (\text{since all } \gamma_n \in \mathbb{R}). \quad (2)$$

This is a Pandrosion polynomial, and *every identity from Papers 1–41 applies*.

2 The six Pandrosion conditions

2.1 Condition 1: Resolvent = logarithmic derivative of ξ

Theorem 2.1. *The resolvent of H is the Pandrosion Cauchy transform:*

$$G(z) = \text{Tr}(zI - H)^{-1} = \frac{P'(z)}{P(z)} = \sum_{n=1}^N \frac{1}{z - \gamma_n}. \quad (3)$$

In the $N \rightarrow \infty$ limit, this must converge to (a normalisation of) $(\xi'/\xi)(1/2 + iz)$.

2.2 Condition 2: Vanishing identity

Theorem 2.2.

$$\sum_{n=1}^N \frac{1}{P'(\gamma_n)} = \sum_{n=1}^N \frac{1}{\prod_{j \neq n} (\gamma_n - \gamma_j)} = 0. \quad (4)$$

Verification: For $N = 30$ zeros: $\sum 1/P'(\gamma_n) = 6.88 \times 10^{-42} \approx 0$.

2.3 Condition 3: Higher Pandrosion zeta vanishing

Proposition 2.3. *For P of degree $N \geq 2s$:*

$$\zeta_P(s) = \sum_{n=1}^N P'(\gamma_n)^{-s} = 0 \quad \text{for } s = 1, 2, \dots, N-1. \quad (5)$$

Verification ($N = 30$):

s	$\zeta_P(s)$
1	6.88×10^{-42}
2	2.32×10^{-83}
3	1.01×10^{-124}

All vanish to machine precision. This is a consequence of the partial fraction identity: the coefficient of z^{-k} in the Laurent expansion of $1/P(z)$ at infinity vanishes for $k = 1, \dots, N-1$ (since P is monic of degree N).

2.4 Condition 4: Density of states

Proposition 2.4. *The spectral density is:*

$$\rho(E) = -\frac{1}{\pi} \operatorname{Im} G(E + i0^+) = -\frac{1}{\pi} \operatorname{Im} \frac{P'(E + i0^+)}{P(E + i0^+)} = \frac{1}{N} \sum_{n=1}^N \delta(E - \gamma_n). \quad (6)$$

The smoothed version must match the Riemann-von Mangoldt formula:

$$N(T) \sim \frac{T}{2\pi} \log \frac{T}{2\pi e} + \frac{7}{8}, \quad \rho_{\text{avg}}(E) \sim \frac{1}{2\pi} \log \frac{E}{2\pi}. \quad (7)$$

2.5 Condition 5: Tidal field = pair correlation

Proposition 2.5. *The tidal field (Paper 26)*

$$T(z) = \frac{E_P(z)}{P(z)^2} = \sum_{n=1}^N \frac{1}{(z - \gamma_n)^2} \quad (8)$$

encodes the two-point correlation of the zeros. For the Hilbert-Pólya operator, the ensemble-averaged tidal field must reproduce the GUE pair correlation:

$$R_2(x) = 1 - \left(\frac{\sin \pi x}{\pi x} \right)^2. \quad (9)$$

2.6 Condition 6: Spectral determinant

Proposition 2.6.

$$\log |D^2| = \sum_{n=1}^N \log |P'(\gamma_n)| = 2 \sum_{i < j} \log |\gamma_i - \gamma_j|. \quad (10)$$

For $N = 30$: $\log |D^2| = 3153.15$. The growth $\log |D^2| \sim c N^2 \log N$ is consistent with GUE eigenvalue statistics.

3 Level repulsion and sign structure

Proposition 3.1. *For the first N zeros of ζ , the values $P'(\gamma_n) = \prod_{j \neq n} (\gamma_n - \gamma_j)$ have **alternating signs** (for small N):*

N	$\zeta_P(1)$	Signs alternate?	$\log D^2 $
5	10^{-17}	✓	41.6
10	10^{-16}	✓	217.7
15	10^{-17}	✓	555.1
20	10^{-23}	<i>partial</i>	1107.7
30	10^{-42}	<i>partial</i>	3153.2

The sign pattern $(-1)^{n-1}$ for $P'(\gamma_n)$ reflects the interlacing of zeros with the critical points of P , which is guaranteed for real roots of any real polynomial.

4 Operator constraints

If the Hilbert–Pólya operator H exists in a finite truncation as an $N \times N$ self-adjoint matrix, its matrix elements in any basis must satisfy:

1. $\text{Tr}(H) = \sum \gamma_n = 1905.77$ (for $N = 30$).
2. $\text{Tr}(H^2) = \sum \gamma_n^2 = 139307.35$.
3. $\text{Tr}(zI - H)^{-1} = P'(z)/P(z)$ for all $z \notin \{\gamma_n\}$.
4. $\sum 1/P'(\gamma_n) = 0$ (vanishing identity).
5. $|D^2| = \prod |P'(\gamma_n)| = e^{3153.15}$ (spectral determinant).

Any candidate operator can be *tested* against these conditions.

5 Conclusion

Pandrosion	Hilbert–Pólya H	Riemann ζ
$P(t) = \prod (t - \gamma_n)$	$\det(tI - H)$	Hadamard product of ξ
P'/P	Resolvent $\text{Tr}(zI - H)^{-1}$	ξ'/ξ
$\sum 1/P'(\gamma_n) = 0$	Spectral constraint	Zero constraint
$E_P = \sum Q_n^2$	Energy of spectrum	Energy of zeros
$T = E_P/P^2$	Pair correlation	Montgomery R_2
$D^2 = \prod P'(\gamma_n)$	Spectral determinant	$\prod (\gamma_i - \gamma_j)^2$

The Pandrosion framework provides a set of *necessary conditions* that any Hilbert–Pólya operator must satisfy. These conditions—resolvent structure, vanishing identity, tidal pair correlation, and spectral determinant growth—are all verified on the first 30 Riemann zeros. They offer concrete, falsifiable tests for candidate operators and connect the Hilbert–Pólya program directly to the Pandrosion quotient $Q(\gamma_n, z) = P(z)/(z - \gamma_n)$.

References

- [1] G. Pólya, *Bemerkung über die Integraldarstellung der Riemannschen ξ -Funktion*, Acta Math. **48** (1926), 305–317.
- [2] M. V. Berry, J. P. Keating, *The Riemann zeros and eigenvalue asymptotics*, SIAM Rev. **41** (1999), 236–266.
- [3] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. **5** (1999), 29–106.
- [4] I. Besevic, *The Pandrosion zeta function*, Preprint (2026).
- [5] I. Besevic, *Pandrosion and quantum physics*, Preprint (2026).
- [6] I. Besevic, *Pandrosion energy and GUE*, Preprint (2026).